PanDiagnosticCardiology

Plain-Language Cheatsheet

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Given **8 blood tests** and **6 deadly chest conditions**, what is the *smallest, cheapest, fastest* combination of tests that can detect *all* of them?

The Problem

A patient walks into a GP clinic with chest pain. It could be **any** of six life-threatening diseases:

#	Pathology	Plain English
1	ACS	Heart attack
2	PE	Blood clot in the lungs
3	Aortic Dissection	The aorta is tearing apart
4	Pericarditis / Myocarditis	Inflammation of the heart lining / muscle
5	Pneumothorax	Collapsed lung
6	Acute Heart Failure	Heart can't pump enough blood

Today doctors test for **one disease at a time** (HEART score for heart attacks, Wells score for PE). Nobody has a principled method to cover **all six at once**.

The Idea

Borrow a mathematical trick from **cancer drug design** (the ALIN framework):

In Cancer (ALIN)

Find the smallest group of drugs that blocks *every* way a tumour can survive.

Here (Diagnostics)

Find the smallest group of blood tests that *detects* every deadly chest condition.

Both are the same math problem: a **"weighted set cover"** (also called a **minimal hitting set**).

The Inputs

8 Candidate Blood Tests

The Coverage Matrix

Each cell = "how good is this test at detecting this disease?" (sensitivity, 0–1). **Bold** = sensitivity ≥ 0.90 (our threshold).

Biomarker	What it detects	Cost	Time	Blood
hs-cTnI (troponin)	Heart muscle damage	€10.00	15 min	20 μL
D-dimer	Blood clots	€7.00	10 min	15 μL
NT-proBNP	Heart strain	€14.00	15 min	15 μL
CRP	Inflammation	€5.00	4 min	5 μL
Copeptin	Stress hormone	€17.50	20 min	50 μL
H-FABP	Early heart damage	€8.00	15 min	10 μL
Myoglobin	Muscle breakdown	€6.50	10 min	10 μL
Procalcitonin	Bacterial infection	€21.00	20 min	20 μL

	hs-cTnI	D-dim	NT-pro	CRP	Cop	HFABP	Myo	PCT
ACS	**0.95**	0.38	0.60	0.50	0.85	0.84	0.75	0.15
PE	0.45	**0.95**	0.60	0.65	0.55	0.48	0.30	0.35
Ao. Diss.	0.28	**0.97**	0.45	0.72	0.68	0.25	0.30	0.20
Peri/Myo	0.82	0.35	0.70	**0.92**	0.55	0.68	0.58	0.40
Pneumo	0.10	0.15	0.20	0.15	0.30	0.05	0.08	0.10
Ac. HF	0.68	0.60	**0.95**	0.55	0.78	0.55	0.40	0.30

Nothing works. Maximum sensitivity is only 0.30 (copeptin). Pneumothorax can only be diagnosed by physical exam or chest X-ray—not blood.

The Equations — Explained

Equation 1 — "Does this test cover this disease?"

$$a_{p,b} = \begin{cases} 1 & \text{if } M_{p,b} \geq \tau \\ 0 & \text{otherwise} \end{cases} \qquad \text{(Threshold Gate)}$$

Look up the sensitivity of test b for disease p in the matrix. If it is at least τ (we use $\tau = 0.90$, i.e. 90%), mark it as "**covered**" (1). Otherwise "**not covered**" (0).

Examples:

- D-dimer for PE: sensitivity $= 0.95 \geq 0.90 \Rightarrow a_{\text{PE, D-dimer}} = 1$ ✓
- CRP for ACS: sensitivity $= 0.50 < 0.90 \Rightarrow a_{\text{ACS, CRP}} = 0$ ×

After applying the threshold, the big matrix collapses to a simple yes/no grid:

	hs-cTnI	D-dimer	NT-proBNP	CRP	rest . . .
ACS	✓	×	×	×	×
PE	×	✓	×	×	×
Ao. Diss.	×	✓	×	×	×
Peri/Myo	×	×	×	✓	×
Pneumothorax	×	×	×	×	×
Acute HF	×	×	✓	×	×

Equation 2 — "What to minimise" (The Objective)

$$\min_{\mathbf{x}} \sum_{b \in \mathcal{B}} \Big[\underbrace{w_{\text{size}}}_{=10} + \underbrace{w_{\text{cost}} \cdot c_b}_{=1\,\times\,\text{cost}} + \underbrace{w_{\text{time}} \cdot t_b}_{=0.5\,\times\,\text{time}} + \underbrace{w_{\text{sample}} \cdot v_b}_{=0.1\,\times\,\text{volume}} \Big] x_b \tag{Objective}$$

where $x_b \in \{0, 1\}$ means "include test b in the panel (1) or not (0)."

For each test you include, you pay a **penalty**. The solver picks the subset of tests with the **lowest total penalty** that still covers all detectable diseases.

Penalty term	What it means	Weight	Why?
$w_{\text{size}} = 10$	Fixed cost per test	10	Fewer tests = simpler
$w_{\text{cost}} \cdot c_b$	Reagent cost (€)	1	Cheaper = better
$w_{\text{time}} \cdot t_b$	Turnaround (min)	0.5	Faster = better
$w_{\text{sample}} \cdot v_b$	Blood volume (μL)	0.1	Less blood = better

Worked example — scoring two tests

$$\text{penalty(D-dimer)} = 10 + 1\times 7 + 0.5\times 10 + 0.1\times 15 = 10 + 7 + 5 + 1.5 = \mathbf{23.5}$$
$$\text{penalty(NT-proBNP)} = 10 + 1\times 14 + 0.5\times 15 + 0.1\times 15 = 10 + 14 + 7.5 + 1.5 = \mathbf{33.0}$$

The solver adds up penalties for all selected tests and finds the set with the minimum total.

Equation 3 — "Every disease must be covered" (The Constraint)

$$\forall\, p \in \mathcal{P}^* : \quad \sum_{b \in \mathcal{B}} a_{p,b} \cdot x_b \;\geq\; 1 \tag{Coverage}$$

For *every* detectable disease p, at least one selected test must cover it.

$\mathcal{P}^* = \{$ACS, PE, Aortic Dissection, Pericarditis, Acute HF$\}$ — the 5 "coverable" pathologies. Pneumothorax is excluded because no test reaches 0.90.

Worked example — constraint for PE

$$\underbrace{a_{\text{PE,hs-cTnI}}}_{=0} \cdot x_{\text{hs-cTnI}} \;+\; \underbrace{a_{\text{PE,D-dimer}}}_{=1} \cdot x_{\text{D-dimer}} \;+\; 0 + 0 + \cdots \;\geq\; 1$$

$\Rightarrow$ **D-dimer must be selected** — it is the *only* test that covers PE.

Putting It All Together

The solver solves this as an **Integer Linear Program (ILP)**:

"Turn on/off each of the 8 blood tests (0 or 1). Find the cheapest/simplest combination where every disease has at least one test that detects it with $\geq 90\%$ sensitivity."

The Answer

Test	Covers	Cost	Time	Blood
hs-cTnI	ACS (heart attack)	€10.00	15 min	20 μL
D-dimer	PE + Aortic Dissection	€7.00	10 min	15 μL
NT-proBNP	Acute Heart Failure	€14.00	15 min	15 μL
CRP	Pericarditis/Myocarditis	€5.00	4 min	5 μL
TOTAL	**5/6 diseases (83.3%)**	**€36.00**	**15 min**	**55 μL**

Pneumothorax is uncoverable — needs stethoscope + chest X-ray.

Key Findings at a Glance

1. D-dimer Is Irreplaceable

- Removing D-dimer loses **2 diseases** (PE + Aortic Dissection) = -33.3% coverage
- Appears in **100%** of 1,000 bootstrap iterations
- No other test comes close for those two conditions

2. Pareto Frontier — "What if I can't afford 4 tests?"

Tests	Panel	Coverage	Cost
1	D-dimer alone	33.3%	€7.00
2	D-dimer + CRP	50.0%	€12.00
3	D-dimer + CRP + hs-cTnI	66.7%	€22.00
4	D-dimer + CRP + NT-proBNP + hs-cTnI	83.3%	€36.00

Each extra test adds *exactly one more disease* for increasing cost.

3. Early Presenters ($<$ 2 Hours from Symptom Onset)

- Troponin is **too slow** for very early heart attacks (sensitivity drops $0.95 \rightarrow 0.70$)
- At $\tau = 0.90$: **no biomarker can replace troponin** for ACS $\Rightarrow$ 3 tests (CRP + D-dimer + NT-proBNP), 66.7%, €26

- At $\tau = 0.85$: H-FABP (early sens. 0.88) covers ACS $\Rightarrow$ 4 tests (CRP + D-dimer + H-FABP + NT-proBNP), 83.3%, €34
- Serial troponin (ESC 0/1h algorithm) is **essential** at $\tau = 0.90$

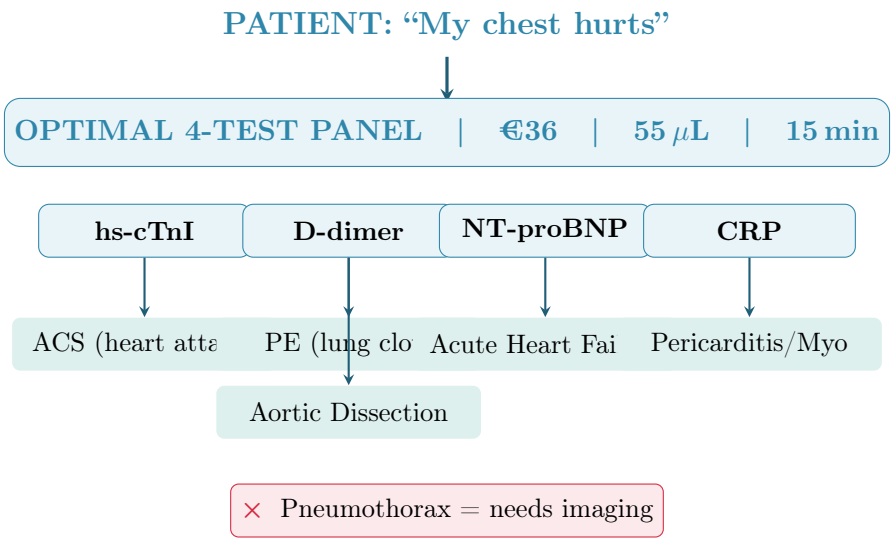
4. Threshold Sensitivity

τ	Tests needed	Note
0.80–0.82	3 tests	CRP drops out
0.85–0.92	4 tests	**Stable regime** (default $\tau = 0.90$)
0.95	3 tests	Coverage drops to 66.7% (loses Pericarditis)

5. Bootstrap Stability (1,000 Iterations)

- D-dimer: selected **100%** of the time
- CRP: selected 78.1% of the time
- hs-cTnI: selected 21.6% of the time
- H-FABP: selected 0.3% of the time
- Mean coverage drops to 50% under CI perturbation

Visual Summary



Key Terminology

SISTER ACT Score: Closing the Gap

The biomarker panel misses pneumothorax (no blood test works). The **SISTER ACT score** fixes this by adding an **AI e-stethoscope** — a digital stethoscope with built-in machine learning that

Term	Meaning
Sensitivity	Probability the test is positive when the disease is truly present (higher = better at detecting)
Specificity	Probability the test is negative when the disease is truly absent (*not modelled here*)
τ (tau)	The sensitivity threshold — test must detect the disease at least this well to "count" (default: 0.90 = 90%)
Coverage	Fraction of the 6 pathologies covered by at least one test above τ
MHS	Minimal Hitting Set — smallest group that "hits" every target
Set Cover	Pick the fewest sets (tests) whose union covers the entire universe (diseases)
ILP	Integer Linear Program — optimisation where variables are whole numbers (0 or 1 = include/exclude)
Pareto optimal	A solution where you can't improve one objective (e.g. cost) without worsening another (e.g. coverage)
Ablation	Removing one component and measuring the damage
Bootstrap	Randomly resample the data many times to test stability
POC	Point-of-care — testing done at bedside/clinic, not sent to a central lab

"listens" for absent breath sounds.

S	Symptoms	0–3	Chest pain typicality, dyspnoea
I	Imaging (AI e-stethoscope)	0–3	Absent sounds (PTX), crackles (HF), rub
S	Signs	0–2	JVD, leg swelling, asymmetric BP
T	Timeline	0–3	< 2h = serial troponin needed
E	ECG	0–3	ST changes, RBBB, S1Q3T3
R	Risk factors	0–3	Age > 65, prior CVD, immobilisation
ACT	Acute Chemistry Testing	0–3	POC biomarker results

Low (0–6): Discharge with safety-netting **Moderate (7–13):** POC serial testing **High (14–20):** Immediate referral

Simulation Results ($n = 10{,}000$)

Metric	HEAR	HEART	SISTER ACT
Sensitivity	96.0%	97.7%	**97.7%**
Specificity	42.4%	49.3%	**70.0%**
NPV (low tier)	.987	.994	**.996**
Missed per 1,000	4.8	2.8	**2.8**
Referral rate	62.2%	56.4%	**38.2%**
PTX sensitivity	93.9%	91.8%	**95.9%**
Pathologies covered	6/6	6/6	**6/6**

Coverage: 5/6 (biomarker only) → **6/6** (+AI e-stethoscope)
Marginal cost: €0.64/patient (device amortised over 3 years)
PTX detection: 14% → **96%** (+82 pp)
Referral reduction: −24 pp vs HEAR, −18 pp vs HEART

The Big Insight

Single-axis testing (one test for one disease) is how we've always done it. But chest pain has 6 possible causes. By borrowing a math trick from cancer drug design, we can find the optimal multi-disease panel with just **4 cheap, fast, fingerprick blood tests**.
This is the same as asking: "What is the minimum number of antibiotics to cover every possible bacterial strain?" — a **covering problem**.

What This Paper Does NOT Do

- × Does NOT use any new patient data — everything comes from published studies
- ✓ **Does** model specificity, PPV/NPV, and panel-level false-positive rate (93.4%)
- ✓ **Does** model serial testing (ESC 0/1h and 0/3h troponin protocols) and time-dependent kinetics
- × Does NOT account for correlations between tests in the same patient
- × Does NOT claim the panel is clinically validated — it is a hypothesis to be tested in the SISTER ACT trial